

Microneedle Superdisintegrant Film Simulator v5

Physics, Mathematics, and Code Architecture

Generated Documentation

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1 Overview

This simulator models drug release from coated microneedles using a hybrid physics–empirical framework combining:

- Geometry-dependent surface area calculations
- Capillary wicking (Lucas–Washburn theory)
- Effective diffusion in porous TPMS structures
- Hydration-controlled film disintegration
- Korsmeyer–Peppas drug release kinetics
- Burst release and sustained release decomposition
- Excipient-dependent transport modifiers

The code is not a first-principles mechanistic model. It is a semi-empirical engineering model calibrated to trends reported in the coated microneedle, oral film, and superdisintegrant literature.

2 Physics Model

2.1 Water Diffusion Through HPMC

The simulator defines an effective water diffusivity:

$$D_{water} = 10^{-10} \left(\frac{5}{\mu_{ref}} \right)^{0.28} e^{-0.03 C_{HPMC}} \quad (1)$$

where:

- μ_{ref} = viscosity grade reference (E5, E15, E50, K4M)
- C_{HPMC} = HPMC concentration (% w/w)

Interpretation:

- Higher HPMC viscosity grade lowers diffusivity.
- Higher polymer concentration lowers diffusivity exponentially.
- Baseline diffusivity is approximately 10^{-10} m²/s.

2.2 Coating Solution Viscosity

The coating viscosity is estimated as

$$\mu = \mu_{ref} \left(\frac{C_{HPMC}}{2} \right)^{2.2} \quad (2)$$

This is an empirical scaling relation.

2.3 Surface Tension

Tween-80 lowers surface tension according to

$$\gamma = 72 - 35 \log_{10} \left(1 + \frac{C_{Tween}}{0.01} \right) \quad (3)$$

bounded between 28 and 72 mN/m.

3 Geometry Module

3.1 Aspect Ratio

$$AR = \frac{H}{R} \quad (4)$$

where:

- H = microneedle height
- R = base radius

3.2 Tip Angle

$$\theta = \tan^{-1} \left(\frac{R}{H} \right) \quad (5)$$

3.3 TPMS Effective Diffusion

For porous geometries:

$$\frac{D_{eff}}{D} = \frac{\varepsilon}{\tau^2} \quad (6)$$

where:

- ε = porosity
- τ = tortuosity

This is a Millington–Quirk style correction.

4 Surface Area Calculations

4.1 Planar Film

$$SA = \pi R^2 \quad (7)$$

4.2 Conical Microneedle

Slant height:

$$s = \sqrt{H^2 + (R - r_t)^2} \quad (8)$$

Surface area:

$$SA = \pi(R + r_t)s \quad (9)$$

where $r_t = 5 \mu m$ is the assumed tip radius.

4.3 Pyramidal Microneedle

$$SA = 4 \times \frac{1}{2}as \quad (10)$$

where $a = 2R$.

4.4 TPMS Structures

The simulator multiplies the external surface area by a geometry-specific surface enhancement factor:

$$SA_{TPMS} = SA_{external} \times SA_{factor} \quad (11)$$

to approximate internal pore surface.

5 Drug Loading

Coating volume:

$$V_{coat} = SA \times L \times f_{vol} \quad (12)$$

Film density:

$$\rho = 1.2 \text{ g/cm}^3 \quad (13)$$

Coating mass:

$$M_{coat} = V_{coat}\rho \quad (14)$$

Drug mass:

$$M_{drug} = M_{coat}f_{drug} \quad (15)$$

6 Capillary Wicking

The simulator implements Lucas–Washburn penetration:

$$x = \sqrt{Ct} \quad (16)$$

with

$$C = \frac{r\gamma \cos \theta}{2\eta\tau^2} \quad (17)$$

where:

- r = effective pore radius
- γ = surface tension
- η = viscosity
- θ = contact angle
- τ = tortuosity

Two separate wicking calculations are performed:

1. Manufacturing coating uptake
2. Interstitial fluid uptake in skin

7 Hydration Kinetics

Hydration rate constant:

$$k_{hyd} = \frac{D_{water}}{L^2} \times 60 \quad (18)$$

modified by:

- Wetting factor
- Superdisintegrant factor
- Gel penalty
- Surface-area-to-volume effects
- TPMS porosity effects

Hydration fraction:

$$H(t) = 1 - e^{-k_{hyd}t} \quad (19)$$

8 Release Model

8.1 Korsmeyer–Peppas

The sustained component follows

$$F = k_{KP}t^n \quad (20)$$

where:

- k_{KP} = release constant
- n = release exponent

8.2 Release Mechanism Interpretation

n	Mechanism
$n \leq 0.45$	Fickian diffusion
$0.45 < n < 0.89$	Anomalous transport
$n \geq 0.89$	Case-II transport

8.3 Burst Release

Burst fraction:

$$F_{burst} = B \left(1 - e^{-k_{burst,eff}t} \right) \quad (21)$$

where

$$k_{burst,eff} = \frac{k_{hyd}k_{burst}}{k_{hyd} + k_{burst}} \quad (22)$$

(modified by gel effects in the implementation).

8.4 Total Release

$$F_{total} = F_{burst} + (1 - B)F_{KP} \quad (23)$$

with clipping between 0 and 100%.

9 Superdisintegrant Effects

9.1 Sodium Starch Glycolate (SSG)

Default parameters:

- $k_0 = 0.11$
- $n_0 = 0.65$
- Burst = 22%
- Gel threshold = 8%

9.2 Croscarmellose Sodium (CCS)

- $k_0 = 0.14$
- $n_0 = 0.55$
- Burst = 27%
- Gel threshold = 5%

9.3 PVPP

- $k_0 = 0.23$
- $n_0 = 0.40$
- Burst = 38%
- Essentially no gel barrier

9.4 Gel Penalty

Above threshold:

$$Penalty = 0.75^{(C - C_{crit})} \quad (24)$$

This reduces hydration and release.

10 TPMS Physics

The simulator contains:

- Gyroid
- Primitive
- Diamond
- Cone + TPMS core

Implicit equations used for visualization:

10.1 Gyroid

$$\sin x \cos y + \sin y \cos z + \sin z \cos x = 0 \quad (25)$$

10.2 Primitive

$$\cos x + \cos y + \cos z = 0 \quad (26)$$

10.3 Diamond

$$\sin x \sin y \sin z + \sin x \cos y \cos z + \cos x \sin y \cos z + \cos x \cos y \sin z = 0 \quad (27)$$

11 Code Architecture

1. User enters formulation and geometry.
2. `materialModel()` computes viscosity, surface tension, diffusivity, and kinetic modifiers.
3. `geometryModel()` computes area, volume, drug loading, TPMS corrections.
4. `simulateWicking()` computes capillary penetration versus time.
5. `simulateRelease()` computes hydration, burst release, sustained release, and total release.
6. Results are rendered using Recharts.
7. TPMS structures are visualized with Three.js.

12 Important Assumptions

- Uniform coating density = 1.2 g/cm³
- Constant diffusivity
- No skin metabolism
- No drug degradation
- No concentration-dependent diffusivity
- Porous structures represented using effective-medium theory
- Korsmeyer–Peppas parameters are empirical calibration constants

13 Likely Literature Sources

The code appears to draw concepts from:

1. Korsmeyer RW et al. Mechanisms of solute release from porous hydrophilic polymers.
2. Peppas NA and Sahlin JJ. Mathematical modeling of drug release.
3. Lucas and Washburn capillary penetration theory.
4. Millington and Quirk tortuosity correction.

5. Microneedle coating and dissolving microneedle literature.
6. HPMC film-forming and oral film formulation studies.
7. CCS, SSG, and PVPP superdisintegrant characterization studies.